If the pole is of higher order, a similar formula holds, one that involves differentiation as well as taking a limit.

Theorem 1.4 *If f has a pole of order n at z_0, then*

$$\mathrm{res}_{z_0} f = \lim_{z \to z_0} \frac{1}{(n-1)!} \left(\frac{d}{dz}\right)^{n-1} (z-z_0)^n f(z).$$

The theorem is an immediate consequence of formula (1), which implies

$$(z-z_0)^n f(z) = a_{-n} + a_{-n+1}(z-z_0) + \cdots + a_{-1}(z-z_0)^{n-1} +$$

$$+G(z)(z-z_0)^n.$$

2 The residue formula

We now discuss the celebrated residue formula. Our approach follows the discussion of Cauchy's theorem in the last chapter: we first consider the case of the circle and its interior the disc, and then explain generalizations to toy contours and their interiors.

Theorem 2.1 *Suppose that f is holomorphic in an open set containing a circle C and its interior, except for a pole at z_0 inside C. Then*

$$\int_C f(z)\, dz = 2\pi i \ \mathrm{res}_{z_0} f.$$

Proof. Once again, we may choose a keyhole contour that avoids the pole, and let the width of the corridor go to zero to see that

$$\int_C f(z)\, dz = \int_{C_\epsilon} f(z)\, dz$$

where C_ϵ is the small circle centered at the pole z_0 and of radius ϵ.

Now we observe that

$$\frac{1}{2\pi i} \int_{C_\epsilon} \frac{a_{-1}}{z-z_0}\, dz = a_{-1}$$

is an immediate consequence of the Cauchy integral formula (Theorem 4.1 of the previous chapter), applied to the constant function $f = a_{-1}$. Similarly,

$$\frac{1}{2\pi i} \int_{C_\epsilon} \frac{a_{-k}}{(z-z_0)^k}\, dz = 0$$

when $k > 1$, by using the corresponding formulae for the derivatives (Corollary 4.2 also in the previous chapter). But we know that in a neighborhood of z_0 we can write

$$f(z) = \frac{a_{-n}}{(z-z_0)^n} + \frac{a_{-n+1}}{(z-z_0)^{n-1}} + \cdots + \frac{a_{-1}}{z-z_0} + G(z),$$

where G is holomorphic. By Cauchy's theorem, we also know that $\int_{C_\epsilon} G(z)\,dz = 0$, hence $\int_{C_\epsilon} f(z)\,dz = a_{-1}$. This implies the desired result.

This theorem can be generalized to the case of finitely many poles in the circle, as well as to the case of toy contours.

Corollary 2.2 *Suppose that f is holomorphic in an open set containing a circle C and its interior, except for poles at the points $z_1, \ldots, z_N$ inside C. Then*

$$\int_C f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

For the proof, consider a multiple keyhole which has a loop avoiding each one of the poles. Let the width of the corridors go to zero. In the limit, the integral over the large circle equals a sum of integrals over small circles to which Theorem 2.1 applies.

Corollary 2.3 *Suppose that f is holomorphic in an open set containing a toy contour γ and its interior, except for poles at the points $z_1, \ldots, z_N$ inside γ. Then*

$$\int_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f.$$

In the above, we take γ to have positive orientation.

The proof consists of choosing a keyhole appropriate for the given toy contour, so that, as we have seen previously, we can reduce the situation to integrating over small circles around the poles where Theorem 2.1 applies.

The identity $\int_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{N} \operatorname{res}_{z_k} f$ is referred to as the **residue formula**.

2.1 Examples

The calculus of residues provides a powerful technique to compute a wide range of integrals. In the examples we give next, we evaluate three improper Riemann integrals of the form

$$\int_{-\infty}^{\infty} f(x)\,dx.$$

The main idea is to extend f to the complex plane, and then choose a family γ_R of toy contours so that

$$\lim_{R\to\infty}\int_{\gamma_R} f(z)\,dz = \int_{-\infty}^{\infty} f(x)\,dx.$$

By computing the residues of f at its poles, we easily obtain $\int_{\gamma_R} f(z)\,dz$. The challenging part is to choose the contours γ_R, so that the above limit holds. Often, this choice is motivated by the decay behavior of f.

Example 1. First, we prove that

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi \tag{2}$$

by using contour integration. Note that if we make the change of variables $x \mapsto x/y$, this yields

$$\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{y\,dx}{y^2+x^2} = \int_{-\infty}^{\infty} \mathcal{P}_y(x)\,dx.$$

In other words, formula (2) says that the integral of the Poisson kernel $\mathcal{P}_y(x)$ is equal to 1 for each $y > 0$. This was proved quite easily in Lemma 2.5 of Chapter 5 in Book I, since $1/(1+x^2)$ is the derivative of $\arctan x$. Here we provide a residue calculation that leads to another proof of (2).

Consider the function

$$f(z) = \frac{1}{1+z^2}\,,$$

which is holomorphic in the complex plane except for simple poles at the points i and $-i$. Also, we choose the contour γ_R shown in Figure 1. The contour consists of the segment $[-R, R]$ on the real axis and of a large half-circle centered at the origin in the upper half-plane.

Since we may write

$$f(z) = \frac{1}{(z-i)(z+i)}$$

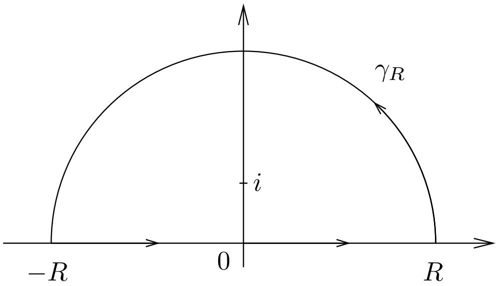

Figure 1. The contour γ_R in Example 1

we see that the residue of f at i is simply $1/2i$. Therefore, if R is large enough, we have

$$\int_{\gamma_R} f(z)\, dz = \frac{2\pi i}{2i} = \pi.$$

If we denote by C_R^+ the large half-circle of radius R, we see that

$$\left| \int_{C_R^+} f(z)\, dz \right| \leq \pi R \frac{B}{R^2} \leq \frac{M}{R},$$

where we have used the fact that $|f(z)| \leq B/|z|^2$ when $z \in C_R^+$ and R is large. So this integral goes to 0 as $R \to \infty$. Therefore, in the limit we find that

$$\int_{-\infty}^{\infty} f(x)\, dx = \pi,$$

as desired. We remark that in this example, there is nothing special about our choice of the semicircle in the upper half-plane. One gets the same conclusion if one uses the semicircle in the lower half-plane, with the other pole and the appropriate residue.

Example 2. An integral that will play an important role in Chapter 6 is

$$\int_{-\infty}^{\infty} \frac{e^{ax}}{1+e^x}\, dx = \frac{\pi}{\sin \pi a}, \qquad 0 < a < 1.$$

To prove this formula, let $f(z) = e^{az}/(1+e^z)$, and consider the contour consisting of a rectangle in the upper half-plane with a side lying

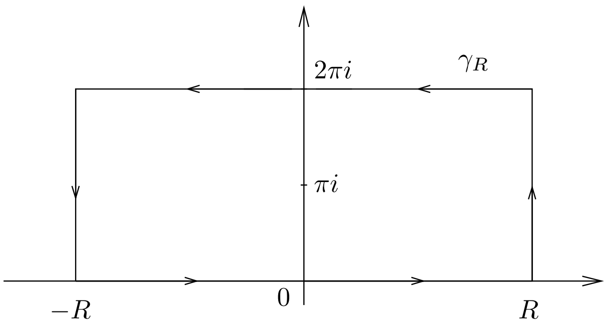

Figure 2. The contour γ_R in Example 2

on the real axis, and a parallel side on the line $\mathrm{Im}(z) = 2\pi$, as shown in Figure 2.

The only point in the rectangle γ_R where the denominator of f vanishes is $z = \pi i$. To compute the residue of f at that point, we argue as follows: First, note

$$(z - \pi i)f(z) = e^{az}\frac{z - \pi i}{1 + e^z} = e^{az}\frac{z - \pi i}{e^z - e^{\pi i}}.$$

We recognize on the right the inverse of a difference quotient, and in fact

$$\lim_{z \to \pi i} \frac{e^z - e^{\pi i}}{z - \pi i} = e^{\pi i} = -1$$

since e^z is its own derivative. Therefore, the function f has a simple pole at πi with residue

$$\mathrm{res}_{\pi i} f = -e^{a\pi i}.$$

As a consequence, the residue formula says that

$$\int_{\gamma_R} f = -2\pi i e^{a\pi i}. \tag{3}$$

We now investigate the integrals of f over each side of the rectangle. Let I_R denote

$$\int_{-R}^{R} f(x)\, dx$$

and I the integral we wish to compute, so that $I_R \to I$ as $R \to \infty$. Then, it is clear that the integral of f over the top side of the rectangle (with the orientation from right to left) is

$$-e^{2\pi i a} I_R.$$

Finally, if $A_R = \{R + it : \; 0 \leq t \leq 2\pi\}$ denotes the vertical side on the right, then

$$\left| \int_{A_R} f \right| \leq \int_0^{2\pi} \left| \frac{e^{a(R+it)}}{1 + e^{R+it}} \right| dt \leq C e^{(a-1)R},$$

and since $a < 1$, this integral tends to 0 as $R \to \infty$. Similarly, the integral over the vertical segment on the left goes to 0, since it can be bounded by Ce^{-aR} and $a > 0$. Therefore, in the limit as R tends to infinity, the identity (3) yields

$$I - e^{2\pi i a} I = -2\pi i e^{a\pi i},$$

from which we deduce

$$\begin{aligned} I &= -2\pi i \, \frac{e^{a\pi i}}{1 - e^{2\pi i a}} \\ &= \frac{2\pi i}{e^{\pi i a} - e^{-\pi i a}} \\ &= \frac{\pi}{\sin \pi a}, \end{aligned}$$

and the computation is complete.

Example 3. Now we calculate another Fourier transform, namely

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \xi}}{\cosh \pi x} \, dx = \frac{1}{\cosh \pi \xi}$$

where

$$\cosh z = \frac{e^z + e^{-z}}{2}.$$

In other words, the function $1/\cosh \pi x$ is its own Fourier transform, a property also shared by $e^{-\pi x^2}$ (see Example 1, Chapter 2). To see this, we use a rectangle γ_R as shown on Figure 3 whose width goes to infinity, but whose height is fixed.

For a fixed $\xi \in \mathbb{R}$, let

$$f(z) = \frac{e^{-2\pi i z \xi}}{\cosh \pi z}$$

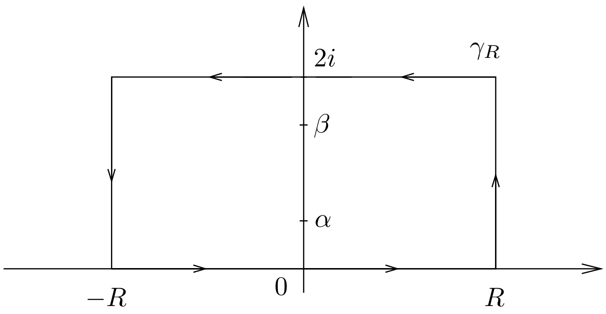

Figure 3. The contour γ_R in Example 3

and note that the denominator of f vanishes precisely when $e^{\pi z} = -e^{-\pi z}$, that is, when $e^{2\pi z} = -1$. In other words, the only poles of f inside the rectangle are at the points $\alpha = i/2$ and $\beta = 3i/2$. To find the residue of f at α, we note that

$$\begin{aligned}(z-\alpha)f(z) &= e^{-2\pi i z\xi}\frac{2(z-\alpha)}{e^{\pi z}+e^{-\pi z}} \\ &= 2e^{-2\pi i z\xi}e^{\pi z}\frac{(z-\alpha)}{e^{2\pi z}-e^{2\pi\alpha}}.\end{aligned}$$

We recognize on the right the reciprocal of the difference quotient for the function $e^{2\pi z}$ at $z = \alpha$. Therefore

$$\lim_{z\to\alpha}(z-\alpha)f(z) = 2e^{-2\pi i\alpha\xi}e^{\pi\alpha}\frac{1}{2\pi e^{2\pi\alpha}} = \frac{e^{\pi\xi}}{\pi i},$$

which shows that f has a simple pole at α with residue $e^{\pi\xi}/(\pi i)$. Similarly, we find that f has a simple pole at β with residue $-e^{3\pi\xi}/(\pi i)$.

We dispense with the integrals of f on the vertical sides by showing that they go to zero as R tends to infinity. Indeed, if $z = R + iy$ with $0 \leq y \leq 2$, then

$$|e^{-2\pi i z\xi}| \leq e^{4\pi|\xi|},$$

and

$$\begin{aligned}|\cosh \pi z| &= \left|\frac{e^{\pi z}+e^{-\pi z}}{2}\right| \\ &\geq \frac{1}{2}\left|\,|e^{\pi z}|-|e^{-\pi z}|\,\right| \\ &\geq \frac{1}{2}(e^{\pi R}-e^{-\pi R}) \\ &\to \infty \qquad \text{as } R\to\infty,\end{aligned}$$

which shows that the integral over the vertical segment on the right goes to 0 as $R \to \infty$. A similar argument shows that the integral of f over the vertical segment on the left also goes to 0 as $R \to \infty$. Finally, we see that if I denotes the integral we wish to calculate, then the integral of f over the top side of the rectangle (with the orientation from right to left) is simply $-e^{4\pi\xi}I$ where we have used the fact that $\cosh \pi\zeta$ is periodic with period $2i$. In the limit as R tends to infinity, the residue formula gives

$$\begin{aligned}I - e^{4\pi\xi}I &= 2\pi i\left(\frac{e^{\pi\xi}}{\pi i}-\frac{e^{3\pi\xi}}{\pi i}\right) \\ &= -2e^{2\pi\xi}(e^{\pi\xi}-e^{-\pi\xi}),\end{aligned}$$

and since $1-e^{4\pi\xi} = -e^{2\pi\xi}(e^{2\pi\xi}-e^{-2\pi\xi})$, we find that

$$I = 2\frac{e^{\pi\xi}-e^{-\pi\xi}}{e^{2\pi\xi}-e^{-2\pi\xi}} = 2\frac{e^{\pi\xi}-e^{-\pi\xi}}{(e^{\pi\xi}-e^{-\pi\xi})(e^{\pi\xi}+e^{-\pi\xi})} = \frac{2}{e^{\pi\xi}+e^{-\pi\xi}} = \frac{1}{\cosh \pi\xi}$$

as claimed.

A similar argument actually establishes the following formula:

$$\int_{-\infty}^{\infty} e^{-2\pi i x\xi}\frac{\sin \pi a}{\cosh \pi x + \cos \pi a}\,dx = \frac{2\sinh 2\pi a\xi}{\sinh 2\pi\xi}$$

whenever $0 < a < 1$, and where $\sinh z = (e^z - e^{-z})/2$. We have proved above the particular case $a = 1/2$. This identity can be used to determine an explicit formula for the Poisson kernel for the strip (see Problem 3 in Chapter 5 of Book I), or to prove the sum of two squares theorem, as we shall see in Chapter 10.

3 Singularities and meromorphic functions

Returning to Section 1, we see that we have described the analytical character of a function near a pole. We now turn our attention to the other types of isolated singularities.